DAY 15 REPORT:

Task 1 – Student Result Table and Box Plot using NumPy, Pandas, and Seaborn.

NumPy, Pandas, and Seaborn are Python libraries used for data handling and visualization. In this task, student roll numbers and marks were created using NumPy arrays. Pass or fail status was generated based on marks. The data was organized into a table using Pandas DataFrame, and a box plot was created using Seaborn to visualize the student marks clearly.

Program:

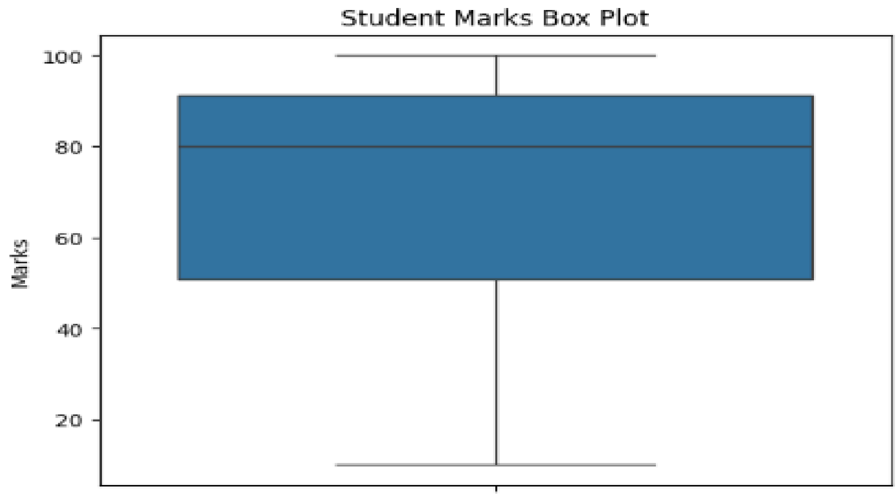
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
roll=np.array([1,2,3,4,5,6,7,8,9,10,11,12])
marks=np.array([85,35,67,25,90,95,56,78,82,10,98,100])
status=np.where(marks>=40,"Pass","Fail")
df=pd.DataFrame({
    "Roll No":roll,
    "Marks":marks,
    "Result":status
})
print(df)
sns.boxplot(y=df["Marks"])
plt.title("Student Marks Box Plot")
```

plt.show()

Output

A table containing roll number, marks, and pass/fail status was created successfully. A box plot was also generated to visualize the marks distribution.

```
OUTPUT
   Roll No  Marks Result
0        1     85   Pass
1        2     35   Fail
2        3     67   Pass
3        4     25   Fail
4        5     90   Pass
5        6     95   Pass
6        7     56   Pass
7        8     78   Pass
8        9     82   Pass
9       10     10   Fail
10      11     98   Pass
11      12    100   Pass
```



Task 2 – Count Plot for Pass and Fail Students

Introduction

In this task, student marks were analyzed to identify pass and fail students. NumPy was used to create arrays, Pandas was used to create a table, and Seaborn count plot was used to visually compare the number of passed and failed students.

Program:

import numpy as np

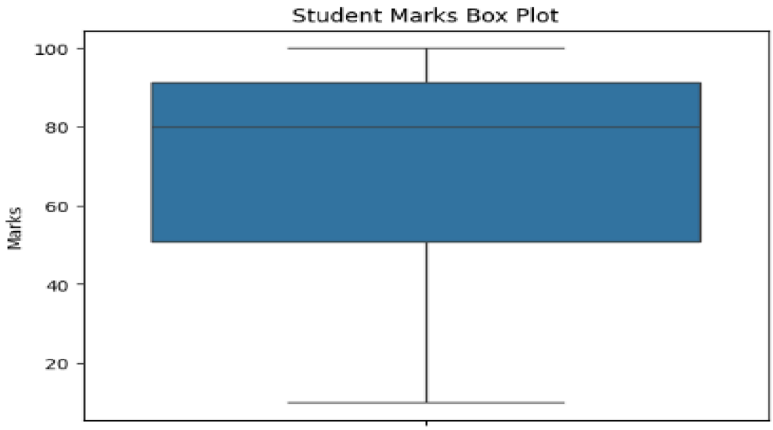
import pandas as pd

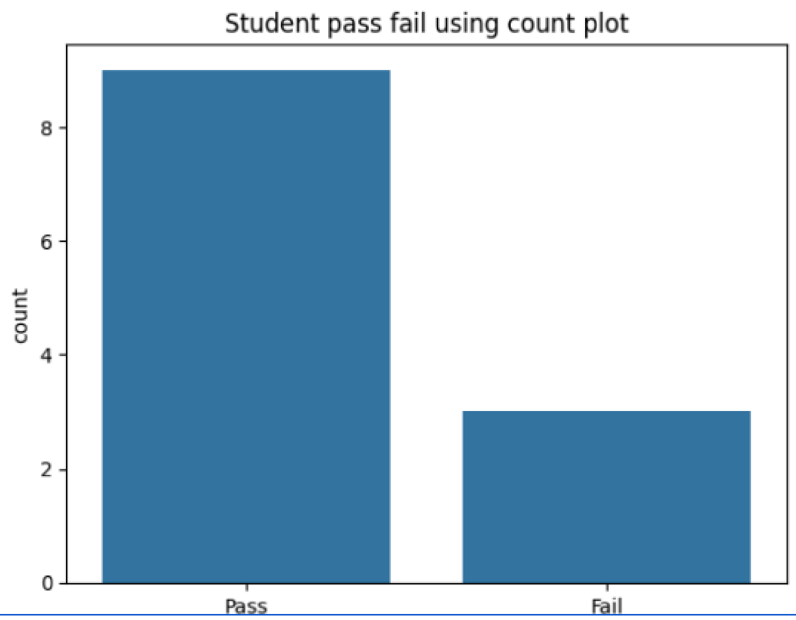
```
import seaborn as sns
import matplotlib.pyplot as plt
roll=np.array([1,2,3,4,5,6,7,8,9,10,11,12])
marks=np.array([85,35,67,25,90,95,56,78,82,10,98,100])
status=np.where(marks>=40,"Pass","Fail")
print("OUTPUT")
df=pd.DataFrame({
    "Roll No":roll,
    "Marks":marks,
    "Result":status
})
print(df)
sns.boxplot(y=df["Marks"])


plt.title("Student Marks Box Plot")
plt.show()
sns.countplot(x=df["Result"])
plt.title("Student pass fail using count plot")
plt.show()
```

Output

A table containing roll number, marks, and result status was displayed. The count plot visually represented the number of pass and fail students.

```
OUTPUT
    Roll No  Marks Result
0        1     85   Pass
1        2     35   Fail
2        3     67   Pass
3        4     25   Fail
4        5     90   Pass
5        6     95   Pass
6        7     56   Pass
7        8     78   Pass
8        9     82   Pass
9       10     10   Fail
10      11     98   Pass
11      12    100   Pass
```





Comparison Graphs in Seaborn

Comparison graphs are used to compare values, categories, or relationships between data.

Graph	Function	Purpose
Line Graph	sns.lineplot()	Compares values over time or sequence
Bar Graph	sns.barplot()	Compares values between categories
Scatter Plot	sns.scatterplot()	Compares relationship between two variables
Box Plot	sns.boxplot()	Compares data distribution
Violin Plot	sns.violinplot()	Compares distribution and density
Count Plot	sns.countplot()	Compares count of categories
Point Plot	sns.pointplot()	Compares category values using points
Swarm Plot	sns.swarmplot()	Compares individual data points
Strip Plot	sns.stripplot()	Compares categorical scatter data

Task 3 – Scatter Plot and Line Plot using Seaborn

Introduction

In this task, student marks and result status were analyzed using Scatter Plot and Line Plot. NumPy was used to create arrays, Pandas was used to organize the data into a table, and Seaborn was used to visualize the comparison between pass and fail students.

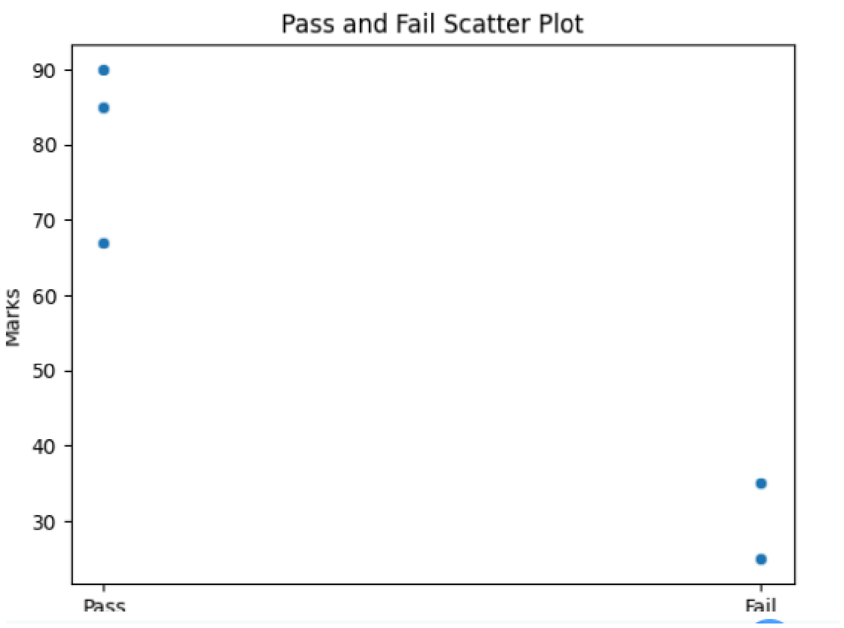
Scatter Plot Program

```
import numpy as np
import pandas as pd
import seaborn as sns
```

```
import matplotlib.pyplot as plt

marks=np.array([85,35,67,25,90])

result=np.where(marks>=40,"Pass","Fail")

df=pd.DataFrame({
    "Marks":marks,
    "Result":result
})

sns.scatterplot(x=df["Result"], y=df["Marks"])

plt.title("Pass and Fail Scatter Plot")

plt.show()
```

Output

A scatter plot was generated successfully to display the relationship between student result status and marks.



2.**Line Graph**

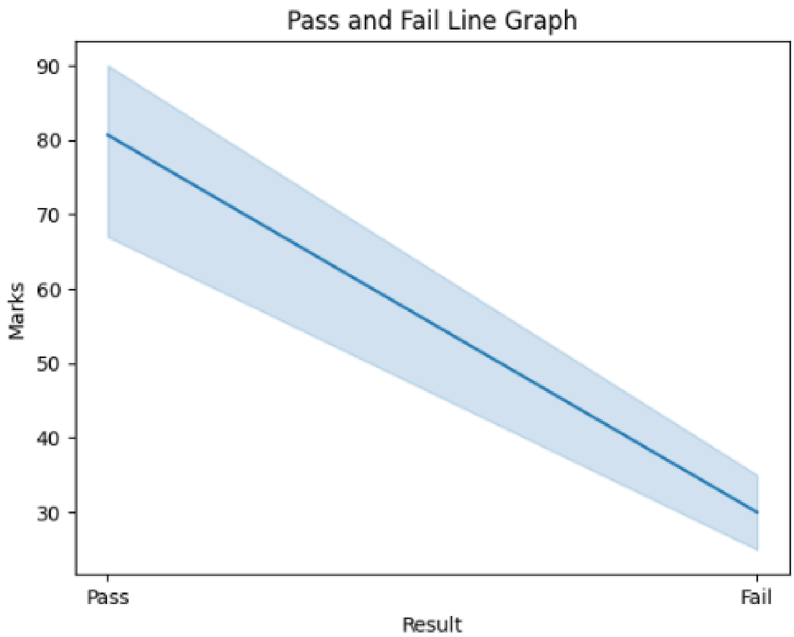
A line graph is used to represent and compare data values using connected lines. It helps to visualize changes and trends between different categories or values.

Program

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

marks=np.array([85,35,67,25,90])

result=np.where(marks>=40,"Pass","Fail")

df=pd.DataFrame({
    "Marks":marks,
    "Result":result
})

sns.lineplot(x=df["Result"], y=df["Marks"])

plt.title("Pass and Fail Line Graph")

plt.show()
```

Output

A line graph was generated successfully to compare the marks of pass and fail students.



Task 4 – Graphs Creation for sample student data using Seaborn

Introduction

Seaborn is a Python visualization library used to create statistical and attractive graphs easily. In this task, an Excel student dataset was read using Pandas and different types of graphs were created using Seaborn to analyze student marks, attendance, and result status visually.

Program

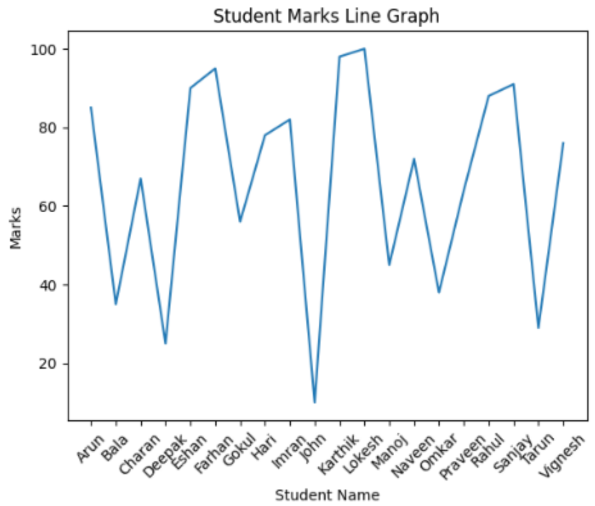
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Reading Excel Dataset
df = pd.read_excel("/content/sampleStudentDataset.xlsx")

# Display Dataset
```

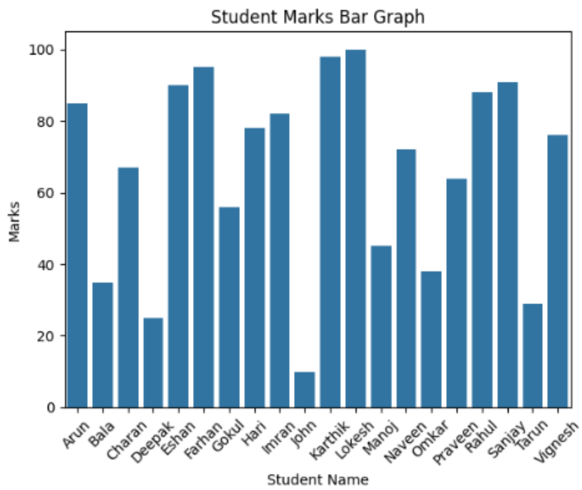
```
print(df)

# 1. Line Graph
sns.lineplot(x=df["Name"], y=df["Marks"])

plt.title("Student Marks Line Graph")
plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.xticks(rotation=45)

plt.show()

# 2. Bar Graph
sns.barplot(x=df["Name"], y=df["Marks"])

plt.title("Student Marks Bar Graph")
plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.xticks(rotation=45)

plt.show()

# 3. Scatter Plot
sns.scatterplot(x=df["Attendance"], y=df["Marks"])

plt.title("Attendance vs Marks Scatter Plot")
plt.xlabel("Attendance")
plt.ylabel("Marks")

plt.show()

# 4. Box Plot
sns.boxplot(y=df["Marks"])
```

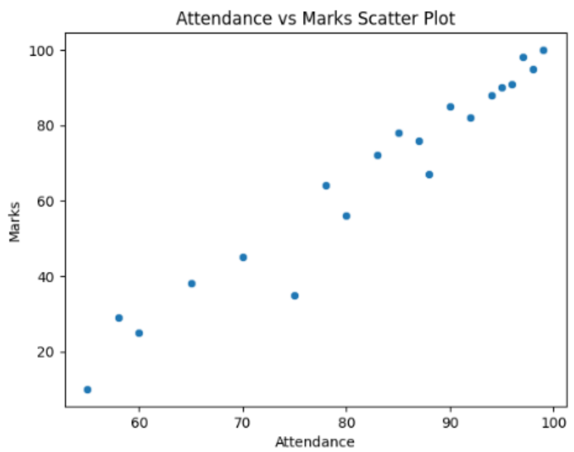
```
plt.title("Student Marks Box Plot")

plt.show()

# 5. Count Plot
sns.countplot(x=df["Result"])

plt.title("Pass and Fail Count Plot")
plt.xlabel("Result")
plt.ylabel("Count")

plt.show()
```

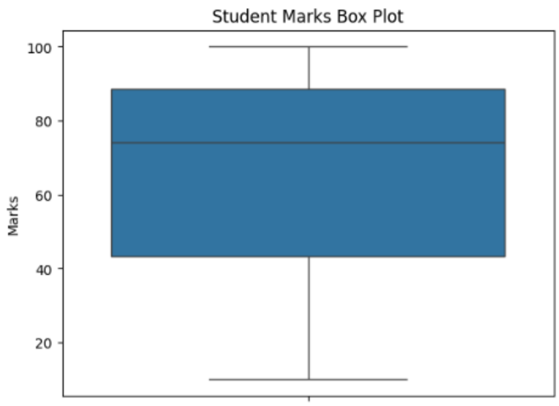
Output

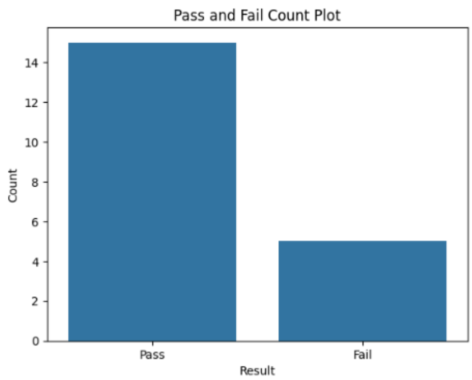
- A line graph was generated to compare student marks.
- A bar graph was created to visualize marks of students.
- A scatter plot was generated to compare attendance and marks.
- A box plot was created to visualize marks distribution.
- A count plot was generated to compare pass and fail student count.


Student Marks Line Graph
Marks
Student Name
100
80
60
40
20
Arun
Bala
Charan
Deepak
Eshan
Farhan
Gokul
Hari
Imran
John
Karthik
Lokesh
Manoj
Naveen
Omkar
Praveen
Rahul
Sanjay
Tarun
Vignesh


Student Marks Bar Graph
Marks
Student Name
100
80
60
40
20
0
Arun
Bala
Charan
Deepak
Eshan
Farhan
Gokul
Hari
Imran
John
Karthik
Lokesh
Manoj
Naveen
Omkar
Praveen
Rahul
Sanjay
Tarun
Vignesh


Attendance vs Marks Scatter Plot
Marks
Attendance
100
80
60
40
20
60
70
80
90
100


Student Marks Box Plot
Marks
100
80
60
40
20


Pass and Fail Count Plot
Count
14
12
10
8
6
4
2
0
Pass
Fail
Result